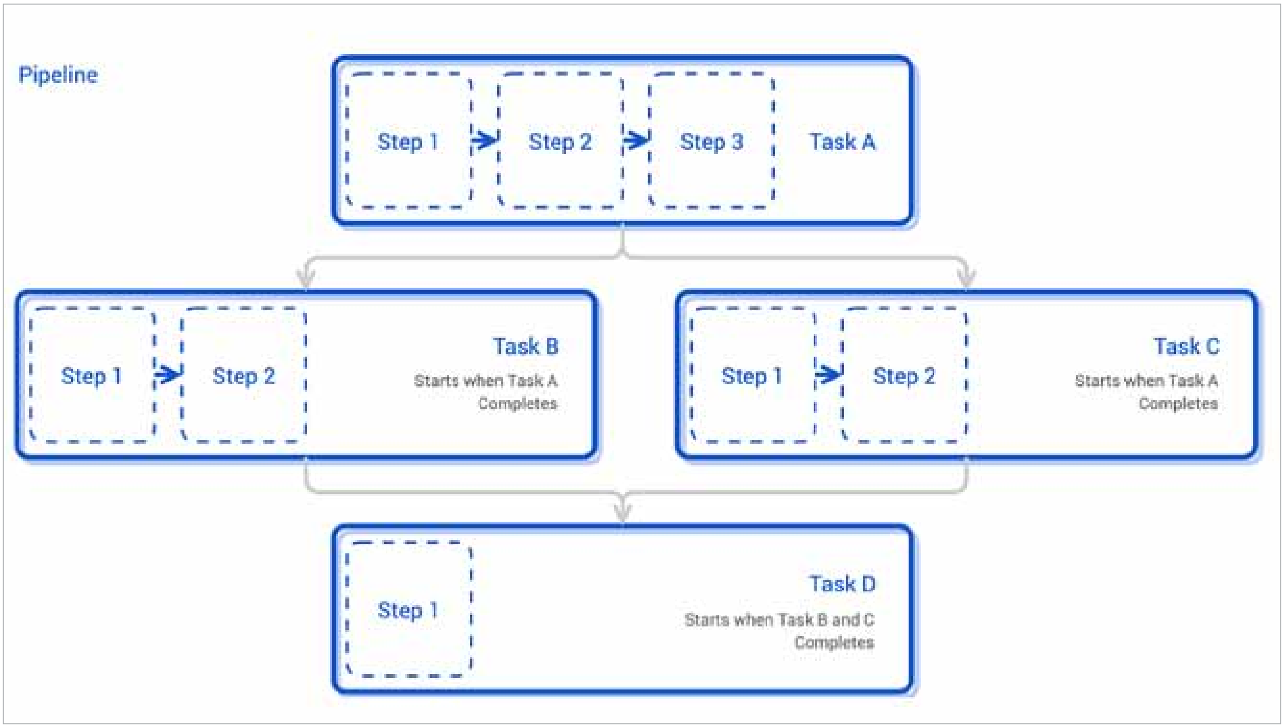


Figure 2: Pipeline

## Anatomy of Tekton Pipelines

Here is a simplified breakdown of Tekton's anatomy.

**Step** is an operation in a CI/CD workflow, such as running some unit tests for a Python web app, or the compilation of a Java program. Tekton performs each step with a container image. Separate containers are created for each step.

At its core, Tekton is designed to simplify the CI/CD process by breaking down complex workflows into a series of smaller, reusable components known as tasks.

**Task** is a collection of steps in order. Tekton runs a task in the form of a Kubernetes pod, where each step becomes a running container in the pod. This design allows you to set up a shared environment for a number of related steps; for example, mounting a Kubernetes volume in a task will be accessible inside each step of the task. Each task will create one pod.

**Taskrun** is a specific execution of a task. Task needs either taskrun or pipelines to get executed. Taskruns can also be used to run a task outside a pipeline.

**Pipeline** is a collection of tasks in order. Tekton collects all the tasks, connects them in a directed acyclic graph (DAG), and executes the graph in sequence. In other words, it creates a number of Kubernetes pods and ensures that each pod completes running successfully as desired. Tekton grants developers full control of the process. You can set up a fan-in/fan-out scenario of task completion, ask Tekton to retry automatically should a flaky test exist, or specify a condition that a task must meet before proceeding.

## Applications of Tekton Pipelines

***Continuous integration:*** Tekton simplifies the process of building and testing code changes with automated CI pipelines. Developers can trigger these pipelines with each code commit, ensuring early detection of issues and promoting a consistent development workflow.

***Continuous deployment:*** Tekton pipelines facilitate the deployment of applications to various environments, such as staging and production. Automated CD pipelines ensure that tested and approved code changes are safely and reliably pushed to production environments.

***Canary and blue-green deployments:*** Using Tekton, teams can implement advanced deployment strategies like canary and blue-green deployments. These strategies allow for gradual rollouts and easy rollbacks, reducing the impact of faulty releases.

***Custom workflows:*** Tekton's flexibility enables teams to create tailored workflows for specific use cases. For instance, they can orchestrate complex multi-step processes involving multiple services and dependencies.

## Tekton pipeline creation

The easiest way of creating a Tekton pipeline is by using a Helm chart, but to